A Survey of Terrestrial and Free Space Based Optical Communications Systems

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Abstract

A low-level operational overview of terrestrial and free space optical communications systems is presented. Communication topics such as modulation, multiplexing, and detection are discussed in relation to optical links. Negative channel effects such as dispersion and absorption are investigated with respect to their impact on the channel and the associated bit error rates. The challenges posed by atmospheric disturbances are considered for free space links. Finally, power link budgets are prepared for both fiber and free space systems to illustrate the losses incurred during transmission.

Introduction & Background

In the development of communications systems throughout history, there are a handful of milestones that standout from the background of progress. Some of these include Marconi's first radio transmission in 1888 and the first communications satellite Anik 1, launched by Canada in 1972. By far one of the greatest breakthroughs in communications was the advent of optical fiber and optical communications through the late 20th century. Fiber cables now span all of the world's continents and oceans linking countries and cultures alike. By now the technology has matured to a have a solid place in our communications infrastructure across the world.

Optical communications (op-com) remains a vibrant research area, and can be divided into two primary groups. The first is fiber based communications systems, which rely on a glass optical fiber as the channel medium. The second is the free-space optical communications system (FSOC), which light that propagating through free space to transmit data. Fiber based systems form the backbone of our communications networks for both voice and data. These networks are robust, fast, and proven. Free space communications on the other hand, have potential to provide fast short to medium distance point to point communications. FSOC networks are very dependent on atmospheric integrity, and near-perfect alignment of the transmitter and receiver. Despite these shortcomings, FSOC networks hold much promise for many different applications.

For their differences, all of the optical networks operate with the same basic components: the transmitter, the receiver, and the fiber/channel. The transmitter contains both the source of the light that will be transmitted and also contains a modulator to encode the data onto the signal. From here, the signal propagates into the channel which will either be an optical fiber for a wired system, or into free space for an FSOC system. At reception, a photodetector senses the incoming photons and demodulates the signal to decode the bits.

The main features and practical implementations behind both fiber and free space optical communication systems will be discussed, keeping in mind the considerations of the network. Ideas such as modulation, detection, and channel response will be examined. The focus will be on the communications aspects and the mechanics of the network.

Fiber Optic Communication Links

Standard optical communications links can span hundreds of miles with no need for amplification or external improvement to signal. For an example of how fast these links are, consider the OC-3 carrier used in network backbones. This link transmits at 155 Mbit/sec, whereas the larger OC-192 carrier can push data at nearly 10 Gbit/sec. The generalized fiber communications system is shown Figure 1 and contains the primary elements [3]

- Modulator and Pre-Signal Processing
- Transmitter
- The fiber channel
- Detector and Post-Signal Processing

Each block is critical in the message delivery. Each of these components will be analyzed in detail.

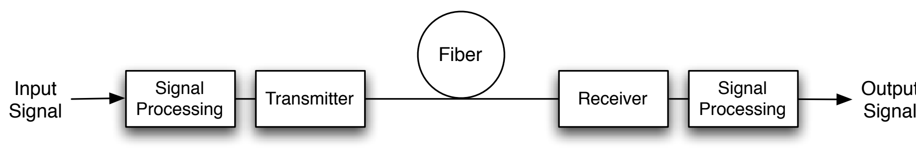

Figure 1: The generalized Fiber Communications Link

Digital Signal Modulation

Essentially, a communications system is a link between two nodes where data is "put" onto a carrier signal and observed at the receiver. How this process is achieved is known as *modulation*. There are several different aspects of the carrier signal that can be altered to rely the bits. This variable may be optical intensity, **E** field amplitude and phase, and even frequency. If the intensity is modulated, the process is referred to as *intensity modulation*; if any aspect of the **E** field is modulated, it is referred to as *field modulation*.

For field modulation, the electric field of the optical carrier is directly changed in accordance with a bit pattern. Generally the phase or amplitude is modulated in accordance with standard digital communications practices. The phase, amplitude, and frequency of the electric field can modulated for communication purposes. An example of amplitude modulation is shown in Figure 2. At first it would seem that field modulation is the obvious mechanism of modulating data for transmission, since it is a natural extension of proven RF methodologies, however there are numerous problems with implementation in the optical frequency range.

Due to the very high frequency of the carrier wave (2×10^{14} Hz at $\lambda = 1.5\mu$m) modulating the electric field is very difficult. First, the source must be of high quality and coherence, free of phase and amplitude irregulaties. Also, because a highly coherent laser is to be used, a single-mode fiber becomes necessary due to the large modal noise exhibited in multi-mode fibers [1]. Because some sort of coherence or understanding is required between the transmitter and the receiver, op-com systems that use field modulation are referred to as *coherent communication systems* and require relatively complicated detection and demodulation systems. These implementations will be discussed later on.

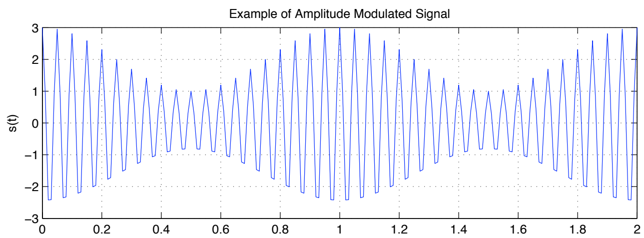

Figure 2: Example of standard amplitude modulation. The message is the low frequency envelope, whereas the carrier is the high frequency sinusoid.

Intensity modulation varies the *intensity* of the carrier in accordance with a bit template by which the signal is coded. This is a much easier system to implement, as the high frequency of the carrier's electric field is unrelated to the process of modulation and demodulation. Nearly all optical communication systems use intensity modulation. Only the power of the carrier is varied at the transmitter, and the differences in power are interpreted as different bits. This can be accomplished by simply varying the input power to the laser diode or LED. The mode of the fiber does not play a roll as it does in field modulation, and the received power can can be measured by a direct-detection receiver.

To encode bits with intensity modulation, each bit is represented by the presence or absence of a pulse of light. This method of encoding is called *on-off keying* (OOK) and is a type of so called line coding. Here, a high value of light is representative of the binary bit "1" and a low value of light is recorded as a binary bit "0". An example of this modulation is shown in Figure 3. This type of OOK is also known as *non-return to zero* encoding (NRZ). In commercial implementations of optical systems, a line code known as carrier-suppressed return to zero (CW-RZ) is used to conserve power and improve inter-symbol interference. For NRZ codes, the sample times needs to be agreed upon by the transmitter and receiver, and timing drift needs to be address. Another type of OOK is return to zero codes (RZ), in which the signal drops to zero between each pulse. This code requires less strict timing (RZ codes are referred to as "self-timing") on the receiver's part but with the trade off of requiring twice the bandwidth as NRZ coding [3], as shown in Figure 4.

Multiple Signal Multiplexing

We would be nowhere close to where we are now in our use of fiber optic networks if we had to send one signal at a time, occupying all of the time on the fiber. To take advantage of the fiber's bandwidth,

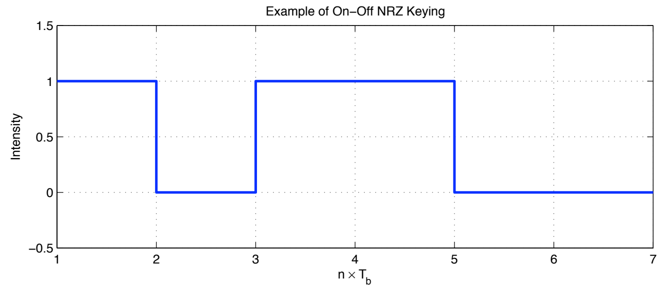

Figure 3: Example of OOK/NRZ Keying. The message sent is 101100.

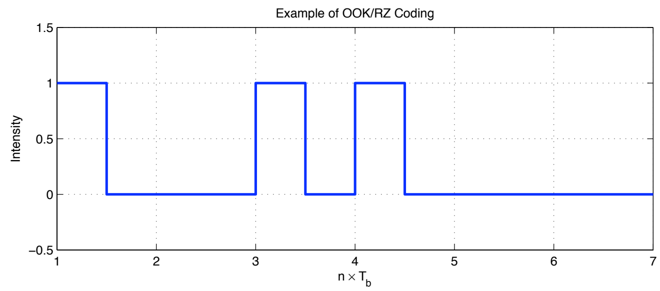

Figure 4: Example of OOK/RZ Keying. The message sent is still 101100.

we need to transmit multiple signals simultaneously. This is accomplished through *multiplexing*. There are a long list of approaches that work very well in the RF analogue of optical communications, including frequency division multiplexing (FDM), time division multiplexing (TDM), and code division multiplexing (CDM). Only some of these are applicable to the optical applications, specifically FDM and TDM.

In optical communications systems using intensity modulation, FDM may be introduced into the system by using optical carriers of different, non-overlapping frequencies. Each optical carrier communicates a distinct message signal. The receiver has optical detectors sensitive to the unique wavelengths of light that are used in transmission, and so the unique data streams are recovered easily. Because the difference in carrier frequency are widely spaced over a few hundred gigahertz, this multiplexing is typically called WDM, or wavelength division multiplexing.

For example, if we consider a 20nm spectral band and $\lambda_0 = 1.55\ \mu$m a frequency spacing of $\Delta f = 250$ GHz is equivalent to a mere 2 nm spacing, so we are able to fill our band with 10 WDM channels. Since we're working with such widely spaced frequencies each channel can be modulated without inter-symbol interference, despite working in a relatively narrow spectral range. At the receiver, a wavelength division demultiplexer uses optical filters tuned to specific frequencies to separate the different transmit wavelengths. These filters can be based on selective absorption, transmission, or reflection [1]. Another detection method is diffractive grading, where the transmission medium is of a graded index, which directs different wavelengths into different fibers for post processing. By allowing multiple WDM channels to simultaneously exist on a single line, we are able to tap into the huge bandwidth of the fiber by spacing our message signals out in frequency. [2]

A different multiplexing technique is TDM. TDM allows a number of different digitized messages or bit streams to timeshare the same the same transmission line. Unlike WDM which has simultaneous propagation of messages, TDM interleaves the bits or groups of bits from different sources just prior to transmission. The receiver has knowledge of the timing setup, and reverses the interleaving to reproduce the distinct messages. TDM is a proven technique in electrical systems – it currently forms the basis of the GSM cell phone system in North America. However in optical systems when it is necessary to multiplex channels operating at or above 10 Gb/sec, the electrical implementations are no longer sufficient. In *optical time-division multiplexing*, a laser beam is split into n message channels, and enters a delay line constructed from a timing shift for each channel. So for instance, the bits for channel one are sent in the first optical time slot, the bits for channel 2 in the second, and so forth. The receiver simply puts everything back together in order. In O-TDM systems, there has to be an even greater agreement on timing than a standard direct-detecton system. O-TDM systems are some of the fastest optical communication systems in use, including systems that operate at 160 Gb/sec.

The Optical Fiber Channel

After we have formatted our data in accordance with some sort of encoding technique and multiplexed the requisite number of signals to take full advantage of our optical fiber, we have to put it onto the transmission line. Though relatively low in dispersion compared to coaxial systems, the optical fiber still has to overcome dispersion issues. In that spirit, there are several things that need to be taken into consideration when evaluating the optical communications link, such as the coupling, the absorption of the fiber, and the scattering that occurs within the fiber.

The first thing that must occur is a coupling with the optical fiber. A coupler directs the light beams that represent the various signals to the appropriate destination. A lot of the coupling is achieved in packaging and cabling. In the simplest coupler, the light emitter is simply butted up against the end of the fiber. Because fibers only capture light within a fairly limited angle, the fiber may not accept all of the optical intensity of the light source. More efficient couplers can be constructed but at the cost of much higher complexity. The expected efficiency of a coupler is something that needs to be evaluated numerically. For now, we will focus on the channel as it relates to the fiber waveguide.

To ensure proper performance, the received intensity must be relatively good to ensure a low bit-error-rate. Absorption can negatively affect the bit-error-rate. Even the purest of glass will have some absorption over some particular wavelengths. This is called *intrinsic absorption*, and is a property of the glass itself. This absorption occurs both in UV and infrared regions of the EM spectrum, and owes its existence to the vibrations of the chemical bonds in the SiO molecule. The UV absorptions occur in a region far removed from where fiber systems operate, so this loss is mitigated. The IR losses can slightly affect the 1.6 μm region of the band, where our systems can operate. Therefore, we note that the absorption losses are mostly insignificant in our regions of interest. If anything, these losses inhibit the extension of fiber systems into UV as well as longer IR wavelengths.

A more debilitating characteristic of the optical fiber is scattering. In the manufacturing process, molecules move randomly through the molten glass. When the glass cools, these perturbations can create localized variations in density that can be modeled as scattering objects in an otherwise smooth material. The optical carrier propagating through the fiber will loose some of it's energy to scattering, called Rayleigh Scattering. This type of scattering is proportional to λ^{-4}, and so it becomes much more important to lower wavelengths. The loss due to Rayleigh scattering is well approximated by the equation

$$L = 1.7(0.85/\lambda)^4$$

where λ is in micrometers and the loss L is in dB/km. Scattering severely limits the ability to communicate with wavelengths below 0.8 μm. To get around the Rayleigh Scattering, either wavelengths of over 2 μm should be used, or a material other than glass is required to propagate. Combining both of these attenuation inducing phenomenon we arrive to the attenuation coefficient, defined in units of dB/km:

$$\alpha = \frac{1}{L} 10 \log_{10} \mathcal{T}^{-1}$$

where $\mathcal{T}$ is the ratio of transmitted power to incident power on a fiber of length L. To mitigate the effects of attenuation on longer lines, optical amplifiers and repeaters would be used to boost signal intensity and reliability.

Bit Error Rate & The Link Budget

The overall performance of any communications system, be it digital or analog is the probability of a bit error, or the bit error rate. What is the chance that if we send a 1, it is decoded a 0 on the other end? A typical BER for an optical system is 10^{-9} (one bit error in a trillion bits), and higher for wireless radio systems (10^{-6}). The BER, as one may suspect, is correlated to the receiver sensitivity. Receiver sensitivity is defined as the minimum number of photons per bit necessary to guarantee that the BER is smaller than the prescribed rate of 10^{-9}. In [1], a proof is offered to show that this number $\bar{n}_0 = 10$ photons per bit corresponds to a BER $\leq 10^{-9}$. This sensitivity corresponds to an optical energy $h\nu\bar{n}_0$ and an optical received power

$$P_r = h\nu \; \bar{n}_0 B_0$$

This power is proportional to the bit rate B_0, and so as the bit rate increases a higher optical power is needed to maintain the proper sensitivity and by proxy, the BER. How else can we quantitatively assess the performance of our optical system? We can prepare a *link budget*, which is a tally of all the power gains and losses incurred in the optical system. In a standard attenuation-limited link, we arrive at the following link budget in dBm units:

$$P_r = P_s - P_c - P_m - \alpha L \quad \text{dB/km}$$

where P_s is the source power of the source, α is the fiber loss in dB/km, P_c is the coupling loss, and L is the fiber length. When P_r is converted to dB, it is evident that P_r increases logarithmically with the data rate B_0. Therefore (and perhaps unsurprisingly), as the bit rate increases the power required to maintain the desired BER also increases. With this in mind, we can derive the maximum length of line L:

$$L = \frac{1}{\alpha}\left(P_s - P_c - P_m - 10 \log \frac{\bar{n}_0 h\nu B_0}{10^{-3}}\right) \quad \text{km}$$

Now using this we can compute the distance versus bit-rate. Basically, this shows the attenuation of the line in bit rate against the distance L [1]. Obviously, this is an important characteristic in an op-com system. Using standard values for $\bar{n}_0$ and $P_c = 0 = P_m$, we can compare this loss on different wavelengths, as seen in the Figure 5.

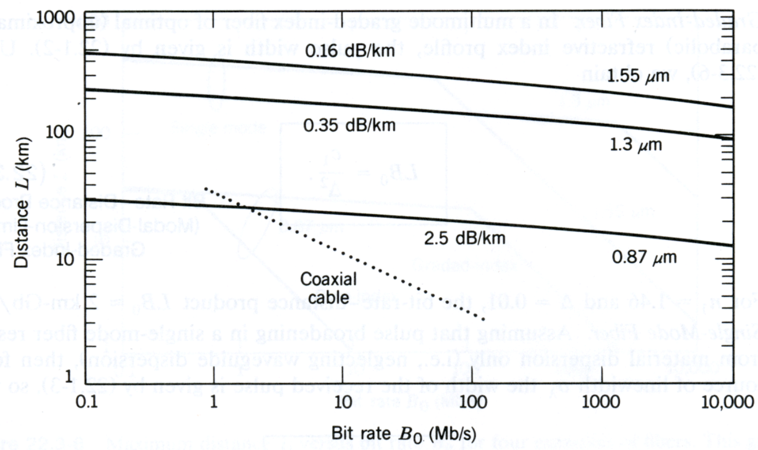

Figure 5: Maximum fiber length L as a function of B_0 in an attenuation-limited fiber. From [1]

After the requisite power level has been determined and engineered at the receiver, the process of decoding and demodulating the beam can begin. At the receiver awaits a photodetector that converts the optical signal to an electrical signal. With OOK, this is a nice and simple process. Numerous different kinds of photodiodes are used to detect and demodulate the light, where the output current of the photodiode is a near replica of the light source at the transmitter. From here we just make the assumption that a high current means a binary 1, and a low current means a binary 0.

In the case of coherent detection when field modulation was used, the process is much more involved. It is possible to obtain the complex amplitude of the optical carrier (both magnitude and phase) by "mixing" it with an oscillator of similar frequency, one whose amplitude and phase are known and coherent. Using a photodetector, it is possible to detect the superposition/interference, and obtain the amplitude and phase this way. This process is called *optical heterodyning*, and increases the receiver's complexity. When the oscillator and the incoming signal have identical frequencies, it is called a homodyne receiver. Typically field modulation can lead to much higher spectral efficiencies than OOK and direct detection. If the receiver can afford the complexity, it may be advantageous to decode this way. Additionally, the heterodyne receiver allows the use of WDM on much frequencies on the order of 100 MHz separation, compared with the 100 GHz of coherent detection. Heterodyne systems are also insensitive to background noisy light that can degrade the efficiency of the direct detector. All of these benefits still come at the cost of the complexity to ensure an optical coupler with perfectly aligned fields, and very sensitive optical phase-locked-loops.

The optical link is very, very efficient in moving large amounts of data a large distance. You can see from the Figure 5 that it single handedly beats the potential of coaxial cable, which is strongly dispersion limited. Combining optical fibers with WDM and optical amplifiers really can fulfill the potential of TB/sec.

The Free Space Optical Channel

In free space optical communications, the propagation channel is free space, be it our own local atmosphere or the vacuum of space. Free space lasers are also a very good alternative when it is either impractical of not feasible to lay fiber line. FSOC systems can be used on commercial skyscrapers to move data between floors at high speeds, and NASA is investigating using FSOC for deep space probe communications. Because of the previously discussed difficulties with frequency and phase modulation, current FSOC systems use intensity modulation with direct detection [5]. Faced in FSOC systems are random fluctuations in the atmosphere's refractive index that can severely degrade the signal over the propagation path, resulting in significantly increased BERs [4]. The primary question in FSOC is how do variations in the atmosphere impact propagation of optical rays?

Atmospheric Losses

As in optical fibers, the two main sources of loss are absorption and scattering. For this case, absorption is caused mainly by water vapor and carbon dioxide, whereas scattering is caused primarily by dust particles and water droplets through fog, rain, or snow. Absorption depends on the water vapor content, and so this depends on the altitude and humidity. Inhomogeneties in the temperature and atmospheric pressure are what causes the difference in refractive index, even in so called "clear sky" conditions. This is what causes the shimmering mirage effects seen on hot days. The two effects which disturb atmospheric transmission most are refraction and *scintillation*, or the random fluctuations in intensity induced by the environment. Because the atmosphere at sea level absorbs some wavelengths better than others, the ranges of $1.5-1.6\mu$m are used. It has been found that scattering has greater variability than absorption, and distinctions need to be made between fog and haze. To understand how the beam looses intensity over distance due to scattering, the following visibility dependent formula was empirically calculated [5].

$$\alpha_{\text{scatter}} = \frac{17}{V}\left(\frac{0.55}{\lambda}\right)^{0.195V}$$

where the visibility V is in km, and λ is in μm. With this equation, it is found that heavy snow and fog most impair the free space link.

It is important and worthwhile to discuss further the trip the beam has to take. For optical systems we are interested in the total received power which is the integrated intensity over the detector area. So the scintillation directly affects FSOC systems. Consequences of this interaction leads to numerous effects, such as an angular deviation of the beam from the line of sight path, resulting in the beam missing the receiver despite near perfect alignment. The link also induces beam spreading, or small angle scattering. This causes a decrease in spatial power density at the receiver, and even a loss in coherence across the wave front.

The Free Space Link

Figure 6 shows the major components of the optical communications link. The transmitter contains the source, a modulator, and has considerations such as coding, bit rate, and amplification. The channel is free-space, and losses incurred come form the aforementioned absorption, scattering, turbulence, and background radiance. It is at the receiver where we demodulate, either through coherent or incoherent detection and decode the bit stream to calculate the bit error rate. The transmitter may be as simple as a

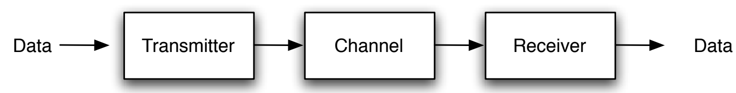

Figure 6: The major components of the simplified optical communications link

diode laser driven with a DC bias, and then modulated using an AC current to provide modulation such as OOK for data transmission. At the receive end, the receive detector (typically a p-i-n photodetector) can be coupled to the receive lens through either an optical fiber or free space. Using a system like this, it is capable to obtain data rates of 10 Gbps with a directly modulated laser, or 40Gbps with a Mach-Zehnder modulator. To align the receive and transmit apertures, a system as crude as a gunsight can be used to something more complex, such as a GPS based system. To assess the total received power and link budget, we use the FSOC link equation:

$$P_{rec} = P_T \left(\frac{d_R^2}{\theta_T^2 L^2} \right) \tau_{atm} \tau_T \tau_R$$

where d_R is the transmit telescope aperture diameter, θ_T is the transmitter beam divergent angle, L is the link range, τ_{atm} is the atmospheric transmission factor, and τ_T is the transmitter optics efficiency.

Determining the reliability and quality of the free space link involves modeling the attenuation coefficient α as a random process whose distribution is empirically defined about a lognormal distribution. It can be shown that the BER for the laser com system in the presence of turbulence is

$$BER = \frac{1}{2} \int_0^\infty p_I(s) \ \mathrm{erfc} \left(\frac{\mathrm{SNR}_s}{2\sqrt{2}(i_s)} \right)$$

assuming OOK with an AWGN channel. From this expression, we see that atmospheric turbulence significantly impacts BER, and as the turbulence and path length increases, as does BER. To successfully decode bits in the presence of the noisy channel, maximum likelihood detection and maximum likelihood sequence detection can be used to mitigate turbulence induced fading [5].

Conclusion

It is clear that our culture has been accustomed to the instant gratification associated with the high bandwidth granted by optical communications. Fiber optic communications are able to deliver us to an age of unprecedented bandwidth, low signal attenuation, small space requirements, and ultimately low cost.

When wired links are not possible due to either space, money, or political reasons the attractive possibility of FSOC exists. FSOC through turbulent channels is under active research and various methods have been developed to mitigate the channel's impact on the bits, and adding more receivers in what is known as spatial diversity can also improve the link.

Fiber op-com links will answer the world's growing communications bandwidth needs, with ever decreasing cost and complexity of implementation. This paper has given a brief overview of the methods of implementation for both free space and wired links, illustrating some of the challenges faced by both systems.

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